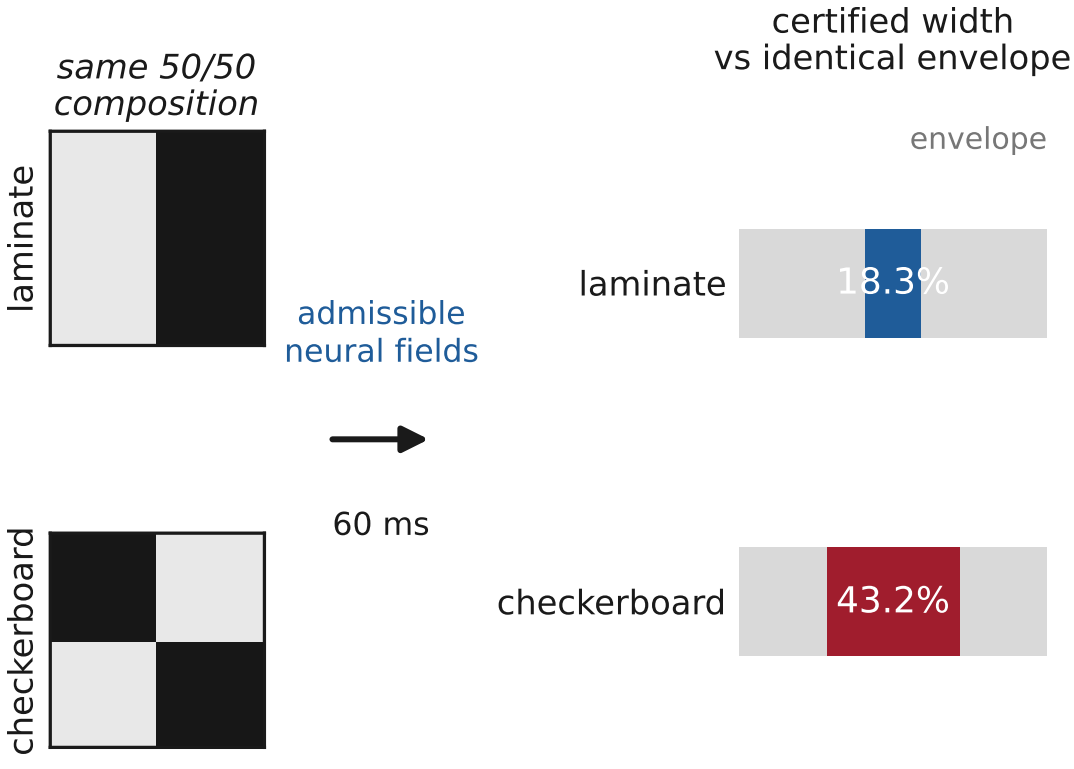

Guaranteed bounds, not estimates

One trained operator certifies microstructures it has never seen, in tens of milliseconds.

The guarantee is architectural: an unfamiliar input widens the interval — it cannot falsify it.

Composition-based bounds give these two structures the same answer. Certified bounds tell them apart.

$K_{\text{low}} \leqslant K_{\text{eff}} \leqslant K_{\text{up}}$ provably, for every input